I Built the Same App
3 Ways on AWS

Which One Actually Makes Sense?

1

2

3

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SETUP

What we're building

A URL Shortener with 3 endpoints

POST **/shorten**
Generate short code, store in DynamoDB

GET **/:shortcode**
Look up code, 301 redirect

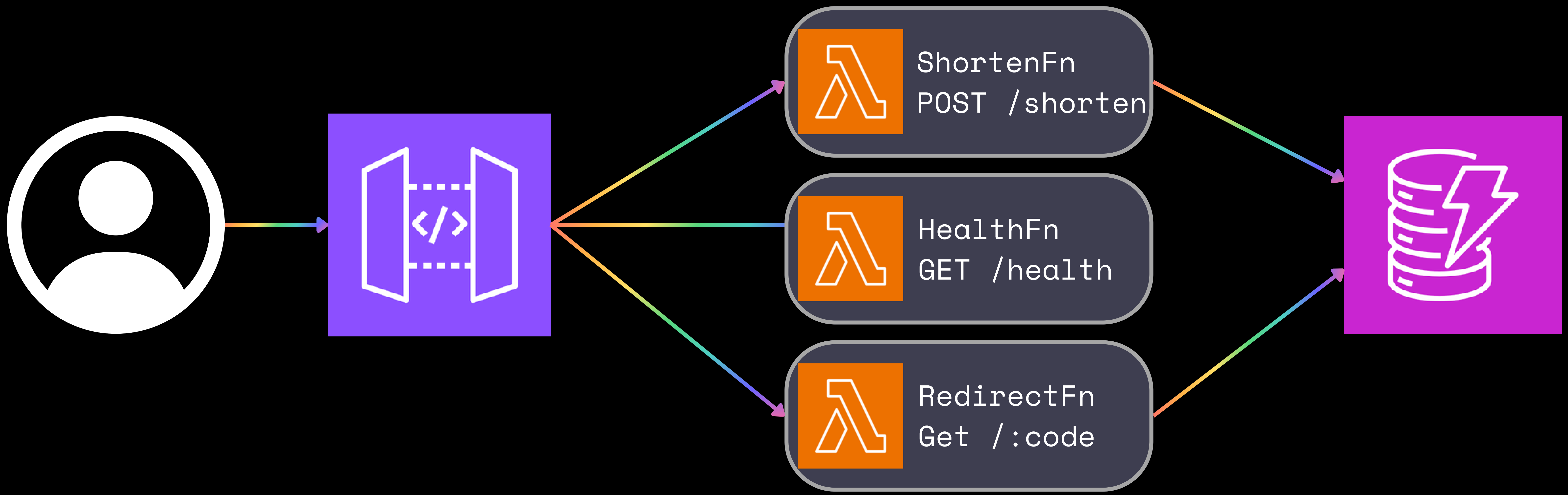
GET **/health**
Returns 200 OK

What stays constant across all 3

- DynamoDB Table - same schema, same SDK calls
- Application code (same logic)
- AWS CDK (Typescript) for all infracture
- BASE_URL env var → returns full short URL

APPROACH 1

Lamba + API Gateway + DynamoDB



APPROACH 1

Reality Check

~200-500ms	~$0/mo	15 min
Cold start - Node.js Lambda	Zero traffic - pay per invocation	Timeout

When it wins

- Variable / unpredictable traffic
- Event-driven workloads
- Cost-sensitive at low volume
- CDK NodejsFunction handles bundling

What to know

- Lambda-specific patterns (not Express)
- Separate handler per endpoint
- New mental model if you know Express
- Context reuse matters for perf

APPROACH 2

EC2 + Express + DynamoDB

APPROACH 2

Reality Check

None	~8-30/mo	None
Cold start - Always-on process	Regardless of traffic	Max timeout

When it wins

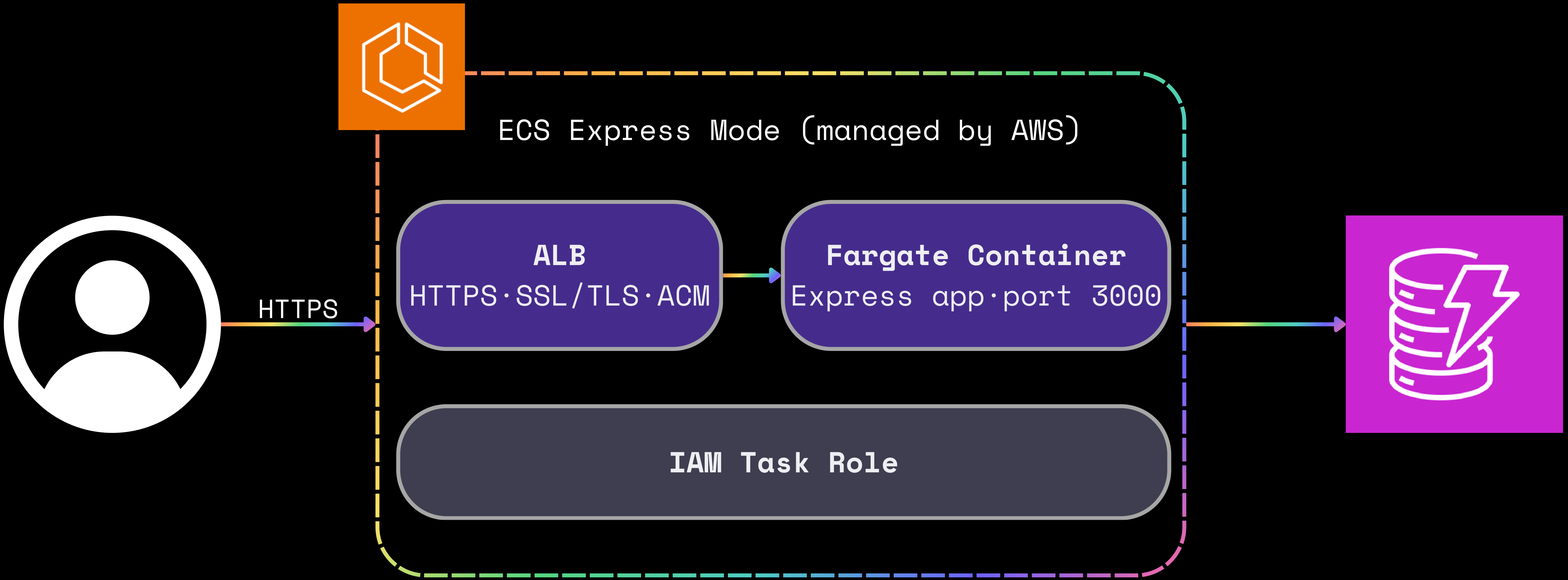
- Familiar Express code – zero rewrite
- Full OS-level control
- Easiest to debug locally
- Long-running or stateful workloads

What to know

- Paying at 3am with zero users
- You manage OS patches + pm2
- Manual scaling (or add Auto Scaling)
- Highest CDK stack complexity

APPROACH 3

ECS Express Mode + Express + DynamoDB



APPROACH 3

Reality Check

None	Min 1	None
Cold start - Fargate tasks always warm	Doesn't Scale to Zero	Max timeout

When it wins

- Standard Express + a Dockerfile
- ALB, HTTPS, and ACM cert are all managed
- Built on Fargate (AWS-managed infrastructure)
- Shares ALB across up to 25 services

What to know

- Docker build step required
- CDK L2 construct still maturing
- Newer service; check docs before recording
- Higher floor cost than Lambda

COMPARE

All three, side by side

	Lambda	EC2	ECS Express
App Code	Lambda handlers	Standard Express	Express + Dockerfile
Cold Start	~200-500ms	None (always on)	None (Fargate warm)
Zero traffic	~$0	~$8-30/mo	Fargate task cost
Auto-scale	Yes - automatic	Manual (or ASG)	Yes - 1 to 5 tasks
Max Time Out	15 min	None	None
You manage	Nothing	OS + pm2 + instance	Container image only

THE VERDICT

Which one should you use?

Traffic variable or unpredictable?	→ Lamba
Existing Express app, minimum rewrite?	→ ECS Express
Need OS-level control or can’t containerize?	→ EC2
Still learning AWS?	→ Build All 3!